Symbiosis Institute of Technology, Pune

Faculty of Engineering
CSE- Academic Year 2026-27
Java Programming – Lab Batch 2024-28

Lab Assignment No:- 22	
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Batch	A1
Class	CSE
Academic Year & Semester	2026-2027/3RD
Date of Submission	07-10-26
Title of Assignment:	To demonstrate and explore the JDBC classes and interfaces PreparedStatement and ResultSet for executing SQL queries and retrieving database records.
Theory:	PreparedStatement is an advanced JDBC interface used to execute precompiled SQL statements with parameters. Unlike the Statement interface, PreparedStatement allows SQL queries to be compiled only once and executed multiple times with different values, thereby improving performance. It also protects applications against SQL injection attacks by separating SQL code from user input. The ResultSet interface stores the records returned by SQL SELECT queries.
Practice Questions	1. Create a login application using PreparedStatement to check username and password.

CODE:

```
import java.sql.*;
import java.util.Scanner;

public class LoginApplication
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Username: ");
        String u = sc.nextLine();
        System.out.print("Password: ");
        String p = sc.nextLine();

        try
        {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/school",
                "root", "abishek@123");

            PreparedStatement ps = con.prepareStatement(
                "SELECT * FROM login " +
                "WHERE username=? AND password=?");
            ps.setString(1, u);
            ps.setString(2, p);

            ResultSet rs = ps.executeQuery();
            if (rs.next())
            {
                System.out.println("Login successful!");
            }
            else
            {
                System.out.println(
                    "Invalid username or password.");
            }
            con.close();
        }
        catch (Exception e)
```

```
        {
            System.out.println(e.getMessage());
        }
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>javac -cp "lib\mysql-connector-j-
26.7.0.jar" LoginApplication.java
C:\Users\Bina Computer\OneDrive\Documents\Java>java -cp ".;lib\mysql-connector-j-
26.7.0.jar" LoginApplication
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Username: abishek
Password: 1234
Login successful!
```

QUESTION	2. Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result.

CODE:

```
import java.sql.*;
import java.util.Scanner;

public class HospitalStaffLogin
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Login ID: ");
        String id = sc.nextLine();
        System.out.print("Enter Password: ");
        String pass = sc.nextLine();

        try
        {
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/school",
                "root", "abishek@123");

            PreparedStatement ps = con.prepareStatement(
                "SELECT role FROM hospital_staff " +
                "WHERE login_id=? AND password=?");
            ps.setString(1, id);
            ps.setString(2, pass);

            ResultSet rs = ps.executeQuery();
            if (rs.next())
            {
                System.out.println("Access granted to "
                    + rs.getString("role"));
            }
            else
            {
                System.out.println(
                    "Invalid Login ID or Password. " +
                    "Access denied.");
            }
            con.close();
        }
        catch (Exception e)
        {
            System.out.println("Error: " + e.getMessage());
        }
        sc.close();
    }
}
```

OUTPUT:

```
C:\Users\Bina Computer\OneDrive\Documents\Java>javac -cp "lib\mysql-connector-j-
26.7.0.jar" HospitalStaffLogin.java
C:\Users\Bina Computer\OneDrive\Documents\Java>java -cp ".;lib\mysql-connector-j-
26.7.0.jar" HospitalStaffLogin
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Enter Login ID: doc101
Enter Password: 12345
Access granted to Doctor
```

Conclusion	The programs successfully demonstrate the use of PreparedStatement and ResultSet in JDBC. Parameterized queries were used to check the login details entered by the user, and the program displayed a success message or the staff role for valid details and denied access for invalid details.